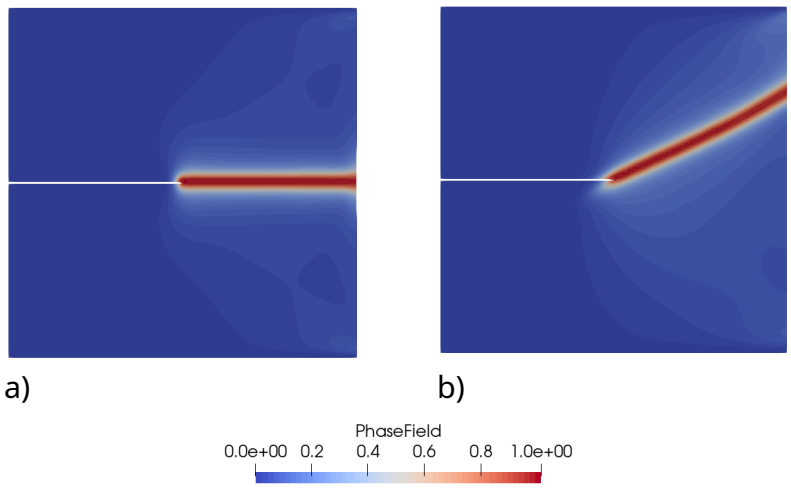
ORTA DOĞU TEKNİK ÜNİVERSİTESİ
MIDDLE EAST TECHNICAL UNIVERSITY

Computational Solid Mechanics Lab (CSML)
Department of Civil Engineering
Middle East Technical University (METU)

Master Thesis (1)

A Phase-Field Method for the Anisotropic Quasi-Brittle Fracture of 3D Printed Concrete : Constitutive Modeling and Calibration

Overview



Representative results for a) isotropic, b) anisotropic fracture using phase-field modeling.

3D printed concrete (3DPC) is rapidly gaining ground in the construction sector owing to an extrusion-based manufacturing process that eliminates the need for formwork. Layer-wise fabrication, however, gives rise to a pronounced fracture anisotropy that governs both the pre-cracking and post-cracking mechanical response, through:

- two mutually orthogonal families of weak interfaces, namely, interlayer and interstrip interfaces
- the extrusion-flow-induced fiber orientation distribution (if fibers are embedded)

Homogenized continuum models in the literature are unable to represent the crack path, crack branching and anisotropic post-peak softening. Besides, interface-based (CZM/XFEM) approaches confine the crack path to predefined surfaces. Discrete (particle-based) models incur enormous computational cost at the structural scale. The aim of this project is to **model the anisotropic quasi-brittle fracture behavior of 3DPC** in a holistic and computationally efficient manner by means of a variational phase-field framework that incorporates print-induced microstructural information. In the proposed method, the interface normal vectors will be reduced to second order structural tensors and embedded in the crack surface density functional. This functional controls the crack direction and the branching observed

between weak interface planes. Subsequently, direction-dependent fracture energy and fracture toughness data reported in the literature will be used for calibration and validation. The expected outcomes include

- a predictive computational tool validated against experimental data
- an analysis capability enabling the direction-dependent safe design of structural 3DPC components

Tasks

- implementing anisotropic user material subroutine based on the identified interface normals and the crack surface density functional
- identifying the elastic constants and the fracture parameters from compression and tension tests loaded along x-, y- and z-directions, as well as from notched three-point or four-point bending tests
- preparation of research findings for journal publications and academic conferences

Your qualifications

- B.Sc. degree holder in Civil Engineering, Mechanical Engineering, or related discipline
- strong motivation to learn continuum and computational solid mechanics
- programming experience (e.g. Fortran, Python) and familiarity with FEA tools are two pluses
- good command of written and spoken English

We offer

- direct research mentorship and guidance throughout the thesis work
- access to computational infrastructure and research facilities
- opportunities for conference participation (contingent on BAP-Career project application)
- scholarship: 12 000 TL/month for 22 months (contingent on BAP-Career project application)

Contact

Interested candidates should send their CV & Letter of Interest to:

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